Predicting Stroke Risk Using Patient Health and Lifestyle Data

DataXplore 2.0 - Stage 01
Data Analysis & Report

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01. Introduction

1.1 Background & Problem Context

Stroke is one of the leading causes of death and disability worldwide, accounting for about 11% of total deaths. This condition occurs when the blood supply to part of the brain is interrupted or significantly reduced, thereby depriving brain tissue of essential nutrients and oxygen. Within minutes, brain cells begin to die, making stroke a medical emergency where early diagnosis becomes the single most important factor influencing the patient's outcome.

Despite considerable progress in medicine, stroke still poses a major burden to public health. According to the World Health Organization (WHO), one out of four people who have reached 25 years of age are estimated to suffer from a stroke at some point in their lives. Stroke is particularly prevalent in low and middle level countries due to the lack of early screening and preventative efforts. Furthermore, even in sophisticated healthcare environments, many strokes happen without any prior indication that a patient is at higher than normal risk.

Modern healthcare systems produce large amounts of data related to patients' characteristics and conditions. Nevertheless, the ability to produce meaningful risk scores based on this data is far from straightforward. While clinical risk scoring methods are often used to estimate stroke risk (e.g., Framingham Stroke Risk Score), their effectiveness is somewhat hampered by the use of only a small subset of variables and assumptions related to linear relationships between variables. Machine learning (ML) methods provides an excellent means of addressing this challenge as it allows for learning data patterns from large datasets, taking care of variable interactions, and producing personalized risk scores.

This data analysis and report aims to bridge that gap by using patient health and lifestyle data to develop, evaluate, and interpret machine learning models for stroke risk prediction, to provide insights that can support clinical decision making.

1.2 Objectives of the Analysis

This comprehensive analysis and report was conducted using the given dataset of 9722 patient records such as demographics, clinical, and lifestyle information, as part of DataXplore 2.0 - Stage 01. This analysis core objectives are:

- Develop and evaluate predictive models for stroke risk classification.
- Identify the most influential clinical and lifestyle factors contributing to stroke risk.
- Generate actionable insights to support preventive healthcare strategies.

1.3 Key Findings (Preview)

XGBoost (eXtreme Gradient Boost) Tuned model achieved the best AUC ROC of 0.8512 and the highest sensitivity of 0.82, catching 82% of actual stroke patients on the test dataset.

Age (mean 67.7 vs 42.0 years, gap = 25.8 years, $p<0.001$), glucose level (132.5 vs 104.8 mg/dL), and BMI are the three strongest predictors, consistent across all 4 models and all statistical tests. Dropping derived redundant features improved AUC by 0.080 over the baseline, demonstrating that feature quality outweighs feature quantity.

02. Methodology

2.1 Dataset Description

The provided healthcare dataset for this analysis and report contained 9722 patient records, which spans across 17 variables that cover four broad categories of patient information. We have identified that Demographic variables include age, gender, and marital status, Clinical measurements include blood glucose level, body mass index (BMI), hypertension status, and heart disease status, Lifestyle indicators include smoking habit, employment type, and residence type and finally, derived risk variables include age group categorization, BMI category, glucose risk flag, lifestyle risk score, and a composite risk score.

Our target variable, stroke_event, is binary coded as 1 if the patient experienced a stroke and otherwise if not 0. This binary classification setup is the foundation of our implementation of the predictive modelling task. The dataset represents a clinical data collection scenario, but not all features are perfectly clean, some are redundant, and the target classes are heavily imbalanced. The dataset was required to handle carefully before modelling could start.

2.2 Tools & Technologies

We conducted this analysis in Python 3.10 version using a cloud based Jupyter notebook environment, Google Colab. Throughout the pipeline, pandas and numpy Python libraries were used to handle all data manipulation, cleaning, and transformation operations. All 19 visualizations were produced using matplotlib and seaborn libraries. Statistical testing was performed using SciPy, specifically the ttest_ind, mannwhitneyu, chi2_contingency, and pearsonr functions. Machine learning models were built and evaluated using scikit-learn, with XGBoost implemented via the dedicated xgboost library. We used the SHAP library for model explainability, which implements SHapley Additive exPlanations. It's a theoretically grounded framework for interpreting machine learning predictions. Since this is a medical to healthcare environment, for the evaluation metrics, we went beyond standard accuracy to include Sensitivity, Specificity, Positive Predictive Value (PPV), Negative Predictive Value (NPV), F1 score, and AUC-ROC all computed using sklearn.metrics.

Component	Tools Used
Programming Language	Python 3.10 (with Google Colab)
Data Manipulation	pandas, numpy
Visualisation	matplotlib, seaborn
Statistical Tests	scipy.stats (ttest_ind, mannwhitneyu, chi2_contingency, pearsonr)
Machine Learning	scikit-learn, xgboost

Explainability	SHAP (SHapley Additive exPlanations)
Evaluation	Sensitivity, Specificity, PPV, NPV, AUC ROC, F1, CV AUC

2.3 Analytical Pipeline Overview

For this analysis, we followed a structured nine step pipeline which we designed to maximize rogor while preventing data leakage at every stage.

- **Step 1 - Data Loading & Audit:** The raw dataset was loaded and profiled to identify data quality issues, including missing values, duplicate records, and potential sources of data leakage. Then we removed the unnamed index column, because it was found to be sorted by the target variable, a direct leakage risk.

- **Step 2 - Deduplication:** A total of 4612 duplicate records were identified and removed, reducing the dataset from 9722 to 5 110 records. This step was important because the duplicate records artificially expanded the dataset and, more importantly, they disrupted the natural class distribution. After deduplication, the true stroke prevalence of 4.87% was restored, revealing the actual severity of class imbalance.

- **Step 3 - Feature Audit:** We evaluated each feature for its clinical relevance and statistical independence. The employment_type variable was identified as a confounded proxy for age (Pearson $r=0.191$, $p<0.001$), with the "other" category comprising almost exclusively children with a mean age of 7.1 years. Additionally, five derived variables (age_group, high_glucose, bmi_category, lifestyle_risk, risk_score) were identified as redundant, all computed from raw features already present in the dataset. Both categories were excluded from the final model feature set.

- **Step 4 - Encoding:** We encoded categorical variables appropriately to make them ready for machine learning. Nominal features with no inherent order (gender, residence, smoking_habit) were one-hot encoding with drop first to avoid the dummy variable trap. Ordered categorical variables (age_group, bmi_category, lifestyle_risk) received ordinal integer mapping that preserved their natural hierarchy.

- **Step 5 - Train/Test Split:** As a critical leakage prevention step, sn 80/20 stratified split was performed before any imputation, scaling, or fitting operations. Stratification ensured the same 4.87% stroke prevalence was maintained in both training (4 088 samples) and test (1 022 samples) sets.

- **Step 6 - Leakage Safe Imputation:** Computing the imputation value from the full dataset before splitting would be a mistake that would have allowed test set information to influence the imputation, constituting data leakage. Therefore, we imputed the missing 201 BMI values using the median BMI computed exclusively from the training set (28.0). This value was then applied to both training and test sets.

- **Step 7 - Exploratory Data Analysis:** Generated 19 figures to comprehensively explore the dataset, covering univariate distributions, bivariate comparisons, correlation

structures, risk zone interactions, and pairwise relationships between key continuous features.

- **Step 8 - Statistical Analysis:** A full suite of statistical tests was applied to quantify and validate observed patterns, parametric t-tests, non-parametric Mann-Whitney U tests, point biserial correlation, chi square tests, Cramér's V association strength, and odds ratios with 95% confidence intervals. To ensure findings were not sensitive to distributional assumptions, both parametric and non parametric tests were used together.

- **Step 9 - Model Training, Tuning & Evaluation:** Trained four machine learning models on the training set, tuned using cross-validation, and evaluated on the held out test set using medical standard metrics. After that, SHAP values were computed for the best model to provide per feature directional explainability.

Step	Phase	Description
1	Data Loading & Audit	Loaded raw 9722 record dataset. Identified missing values, duplicates, and leakage risks
2	Deduplication	Removed 4612 duplicate records. Restored natural 19.5:1 class imbalance
3	Feature Audit	Identified and dropped employment_type (age proxy, r=0.191). Dropped derived redundant features for modelling
4	Encoding	One-hot for nominal. Ordinal mapping for ordered categories. train/test split before all fitting
5	Imputation	BMI imputed using train only median (28.0) applied to test set leakage safe
6	EDA & Visualisation	19 figures covering distributions, correlations, risk zones, scatter plots, SHAP
7	Statistical Analysis	T-tests, Mann Whitney U, Chi square, Cramér's V, Odds Ratios with 95% CI
8	Model Training	LR, RF, XGBoost, XGBoost Tuned - class_weight / scale_pos_weight balancing
9	Evaluation	Medical metrics, GridSearchCV (162 configs), threshold analysis, 5-fold CV, SHAP

2.4 Class Imbalance Strategy

We have identified a severe class imbalance with only 4.87% stroke cases, a 19.5:1 ratio of non stroke to stroke patients. This imbalance is clinically realistic (because strokes are genuinely rare events in the general population), but this creates a significant modelling challenge, because a naive classifier that predicts "no stroke" for every patient would achieve 95.1% accuracy while being completely useless for screening purposes.

To address the class imbalance, we considered two commonly used approaches, resampling (SMOTE - Synthetic Minority Oversampling Technique) and algorithmic balancing (class weights). Our analysis deliberately chose algorithmic balancing for the following reasons. Generally, SMOTE generates synthetic minority class samples by interpolating between existing minority samples. When we applied it before the train/test split, a common implementation error is that synthetic samples derived from the same original patient record can appear in both training and test sets, causing artificially augmented evaluation metrics. Even when applied correctly after splitting, SMOTE can create unrealistic synthetic patients by interpolating across clinically distinct sub groups.

Instead, class_weight = "balanced" was applied to Logistic Regression and Random Forest, causing the algorithm to penalise misclassification of the minority class proportionally to its scarcity. For the XGBoost model, scale_pos_weight = 19.54 was computed from training labels only and passed directly to the model, achieving the same effect. This approach is leakage free, requires no synthetic data generation, and is widely recommended in a clinical prediction environment.

2.5 Data Leakage Prevention

Data leakages occur when information from outside the training dataset is used to build or evaluate a model, producing optimistic metrics that will not replicate in real world deployment. In applied machine learning it is one of the most common and consequential errors that happens. Our analysis implemented leakage prevention at every stage of the pipeline.

We have split the train/test datasets as the very first preprocessing step, before any fitting operations. All BMI imputation, StandardScaler fitting, scale_pos_weight computation subsequent transformations were derived from training data and applied separately to the test set. GridSearchCV cross-validation for hyperparameter tuning was performed entirely within the training set, with the test set reserved as a completely unseen holdout for final evaluation. Inside the Logistic Regression pipeline object, the StandardScaler was embedded to ensure it was refitted within each cross-validation fold rather than fitted once on the full training set, which would constitute within CV leakage.

03. Data Analysis

3.1 Data Understanding & Exploration

The initial data understanding step showed several important characteristics of the raw dataset. The dataset contains 9722 records and 17 columns, with data types of integers, floats, and object (string) categories. In our review, we have identified that all 17 features were present for all records except bmi_value, which contained 201 missing values all belonging specifically to patients categorised as "obese" in the bmi_category column. This non random missingness pattern is clinically plausible, because obese patients may have had BMI estimated categorically rather than measured precisely. It also means simple random imputation would be inappropriate if the missing values represent a specific patient subgroup, not a random sample of the population.

Our duplicate record check revealed 4612 duplicate rows 47.4% of the raw dataset. This abnormal high duplication rate suggests the data was previously augmented, possibly using a resampling technique applied without proper deduplication. Keeping these duplicates would have considerably distorted the class distribution, inflated model performance metrics, and violated the statistical independence assumption underlying all subsequent tests.

The target variable, stroke_event, showed a distribution of 4861 no stroke cases (95.13%) and 249 stroke cases (4.87%) after deduplication with a class imbalance ratio of 19.5:1. This imbalance is the defining challenge of the entire analysis and informs every methodological decision from model selection to evaluation metric choice.

➢ **Dataset Overview**

Attribute	Value
Raw records	9722
Features	17
Missing values	bmi_value: 201 records in "obese" category only (non random pattern)
Duplicate records	4612 (47.4% of raw data)
Records after cleaning	5110
Training set (80%)	4088 samples
Test set (20%)	1022 samples
Stroke cases (after cleaning)	249 (4.87%)

No stroke cases	4,861 (95.13%)
Class imbalance ratio	19.5 : 1
Features used in model	16 (after encoding & proxy removal)

Table 1: Raw dataset overview and post cleaning statistics

3.2 Data Preparation

To develop a clinically robust and leakage free predictive model, we implemented a strict data preparation pipeline.

3.2.1 Deduplication

A total of 4612 duplicate records were removed using pandas drop_duplicates(). It reduced the dataset from 9722 to 5 110 records. The post deduplication stroke prevalence of 4.87% represents the actual underlying rate in this patient population and is consistent with stroke prevalence estimates in comparable clinical datasets.

3.2.2 Missing Value Imputation

Using the median BMI from the training set (28.0 kg/m²), we imputed the 201 missing BMI values. Median imputation was chosen over mean imputation because BMI distributions in clinical datasets are typically right skewed due to extreme high values. The imputation value was computed exclusively from training data and applied separately to the test set, preventing any form of data leakage. All missing BMI records belonging to the "obese" category were noted but did not change the imputation strategy, as the obese category median (34.7) would have introduced inconsistency with the overall distribution used by the model.

3.2.3 Feature Audit and Selection

We performed a systematic audit on all 17 features to assess clinical relevance, statistical independence, and potential for confounding. The employment_type variable was flagged as a confounded proxy, Pearson correlation with age was $r=0.191$ ($p<0.001$), and the "other" category had a mean age of 7.1 years compared to 50.9 years for government employees and 48.7 years for working adults. This means employment_type was primarily encoding whether a patient was a child or an adult, so it’s a demographic proxy, not a clinical risk factor. Including it would have given the model a disguised age variable, inflating the apparent importance of employment_type and undermining clinical interpretability.

age_group, high_glucose, bmi_category, lifestyle_risk, and risk_score, all these five derived variables were also excluded from the model feature set, because, Each of these is mathematically derived from raw variables already present in the dataset (age, glucose_level, bmi_value). Including both a raw variable and its categorical derivative introduces redundancy

and multicollinearity effectively giving the model the same information twice in different forms. In tree based models, this causes importance scores to be split across the correlated pair, understating the true importance of each. Removal of these five derived features produced a considerable improvement in AUC of 0.080 from 0.767 to 0.839 confirming that these redundant variables were actively degrading model performance.

3.2.4 Encoding Strategy

Gender, residence, and smoking_habit are nominal categorical variables with no natural ordering. Using one-hot encoding we encoded with *drop_first = True* to avoid the dummy variable trap, a form of perfect multicollinearity that destabilises linear models. Age_group, bmi_category, and lifestyle_risk are ordinal variables with a natural hierarchy (young < middle < senior, underweight < normal < overweight < obese, low < medium < high). These were mapped to integer values (0, 1, 2, 3) preserving their natural order, which is more informative for both linear and tree based models than arbitrary one-hot encoding.

Variable Type	Variables	Encoding
Nominal categorical	gender, residence, smoking_habit	One-hot (drop first)
Ordinal categorical	age_group, bmi_category, lifestyle_risk	Ordinal integer mapping
Binary clinical	has_hypertension, has_heart_disease, high_glucose, marital_status	Kept as 0/1
Dropped (proxy)	employment_type	Excluded - age proxy (r=0.191)
Dropped (derived)	age_group, high_glucose, bmi_category, lifestyle_risk, risk_score	Redundant (Derived from raw features)
Continuous	age, glucose_level, bmi_value	Standard Scaler inside LR pipeline only

Table 2: Feature Encoding & Selection Strategy

3.3 Descriptive Statistics

Variable	Overall Mean	Stroke Mean	No Stroke Mean	Median (Stroke)	Median (No Stroke)	Std (Stroke)
Age (years)	43.2	67.7	42.0	71.0	43.0	12.7
Glucose (mg/dL)	106.2	132.5	104.8	105.2	91.5	61.9
BMI	28.9	30.5	28.8	29.7	28.0	6.3
Risk Score	0.15	0.62	0.13	-	-	-

Table 3: Numerical Stats (age, glucose, BMI, risk score)

Variable	Category	Stroke Rate	Key Insight
Hypertension	Yes	13.25%	3.3 times higher than no hypertension (3.97%)
Heart Disease	Yes	17.03%	4.1 times higher than no heart disease (4.18%)
Age Group	Senior	13.15%	Compared to young adults (0.13%)
BMI Category	Obese	6.51%	2.3 times higher than normal BMI (2.82%)
Smoking	Ex-smoker	7.91%	Highest among all smoking categories
Gender	Male	5.11%	Marginally higher than female (4.71%)

Table 4: Stroke Rate by Categorical Variables

Our Descriptive statistics revealed differences between stroke and non stroke patient groups across all key numerical variables. The most striking finding is the age difference, stroke patients have a mean age of 67.7 years and median age of 71.0 years, compared to 42.0 years mean and 43.0 years median for non stroke patients, a gap of 25.8 years in means and 28.0 years in medians. This separation is both statistically dramatic and clinically intuitive, as cumulative vascular damage, arterial stiffness, and comorbidity burden all increase substantially with age.

Glucose level shows the second largest separation, stroke patients average 132.5 mg/dL compared to 104.8 mg/dL for non stroke patients a difference of 27.7 mg/dL. A fasting glucose of 126 mg/dL or above is the clinical threshold for diabetes diagnosis, meaning the average stroke patient in this dataset falls in the diabetic range. Chronically elevated glucose damages blood vessel walls, promotes atherosclerosis, and increases clot formation risk for all direct stroke mechanisms.

BMI shows a smaller but statistically significant difference of 30.5 (stroke) vs 28.8 (no stroke). Both values fall in the overweight to obese range, consistent with the demographic profile of a middle aged to elderly patient population. The smaller absolute difference for BMI compared to age and glucose reflects its role as a contributing rather than primary risk factor.

Among categorical variables, the stroke rate by hypertension status (13.25% vs 3.97%) and heart disease status (17.03% vs 4.18%) represent the most clinically alarming findings. Patients with heart disease experience stroke at more than four times the rate of those without the highest categorical risk differential in the entire dataset. Senior aged patients show a 13.15% stroke rate compared to 0.13% for young patients, a 100 fold difference that underscores the age-risk relationship observed in the numerical analysis.

3.4 Visualizations

A total of 19 diagrams were generated using matplotlib and seaborn libraries throughout the analysis process and were divided into three main groups, exploratory data analysis (Figures 1 to 9), statistical association analysis (Figure 10), and model evaluation and explainability (Figures 11 to 19). Each and every figure was designed specifically for an analytical task or to answer an analytical question rather than simple data visualization.

3.4.1 Stroke Distribution

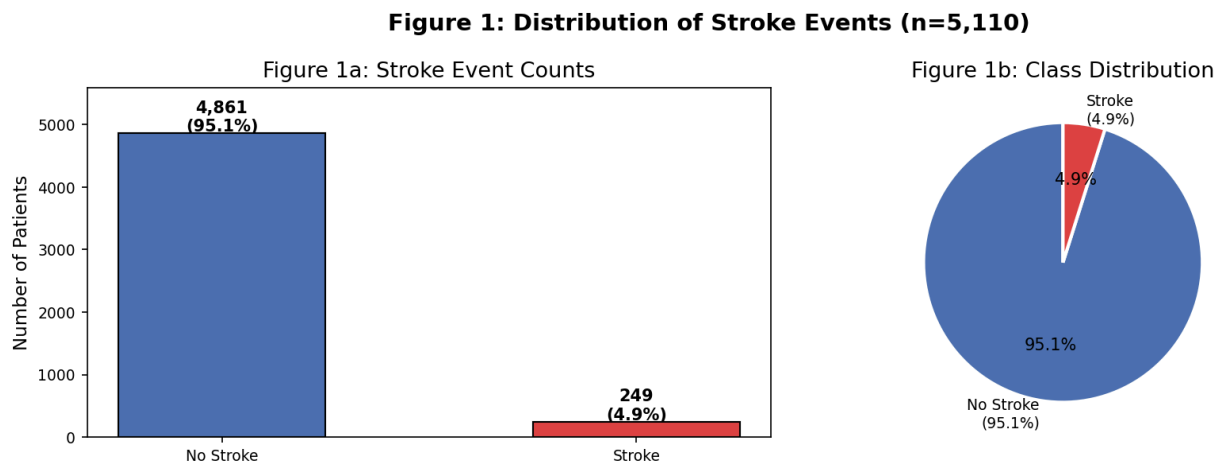

Figure 1: Stroke Distribution (Bar + Pie Chart)

The figure presents the stroke event distribution as both a bar chart and a pie chart. The bar chart allows precise count comparison (4861 vs 249) while the pie chart communicates the proportional imbalance (95.1% vs 4.9%) more intuitively. Together they establish the class imbalance challenge that motivates every subsequent methodological decision.

3.4.2 Stroke Rate by 6 Risk Categories

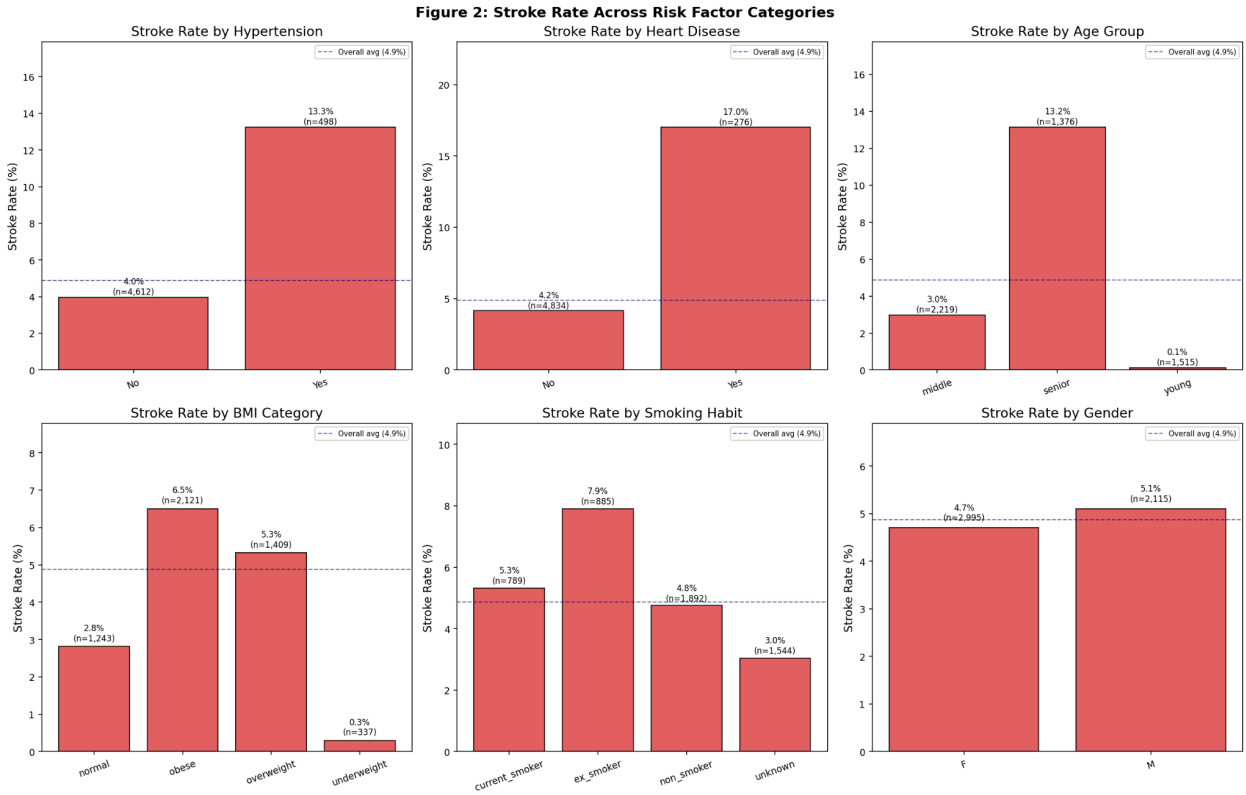

Figure 2: Stroke Rate by 6 Risk Categories

This figure presents a six panel bar chart showing stroke rates across six categorical risk factors simultaneously. The reference line in each panel shows the overall average stroke rate (4.87%), making it immediately clear which categories carry above average risk. Hypertension and heart disease panels show the most dramatic elevations, while gender panels show minimal separation, a finding later confirmed statistically (p=0.360).

3.4.3 Age Distribution

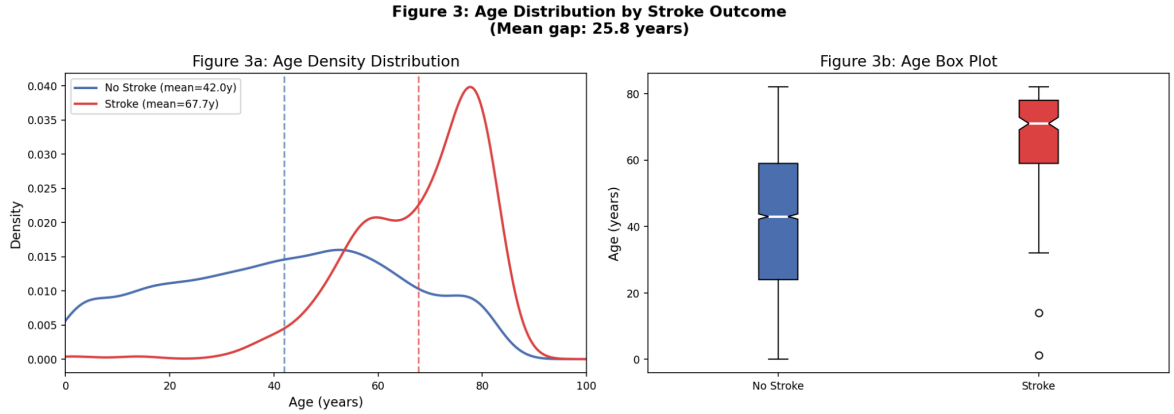

Figure 3: Age Distribution (KDE + Boxplot)

This figure uses a kernel density estimate (KDE) plot and a notched boxplot side by side to visualize the age distribution by stroke outcome. The KDE shows the shape of the distribution, stroke patients form a tight cluster between 65 and 85 years while non stroke patients are more uniformly distributed from childhood to old age. The notched boxplot shows that the notches (representing 95% confidence intervals around the median) do not overlap, confirming that the median age difference is statistically significant without requiring a formal test.

3.4.4 Glucose & BMI Distributions

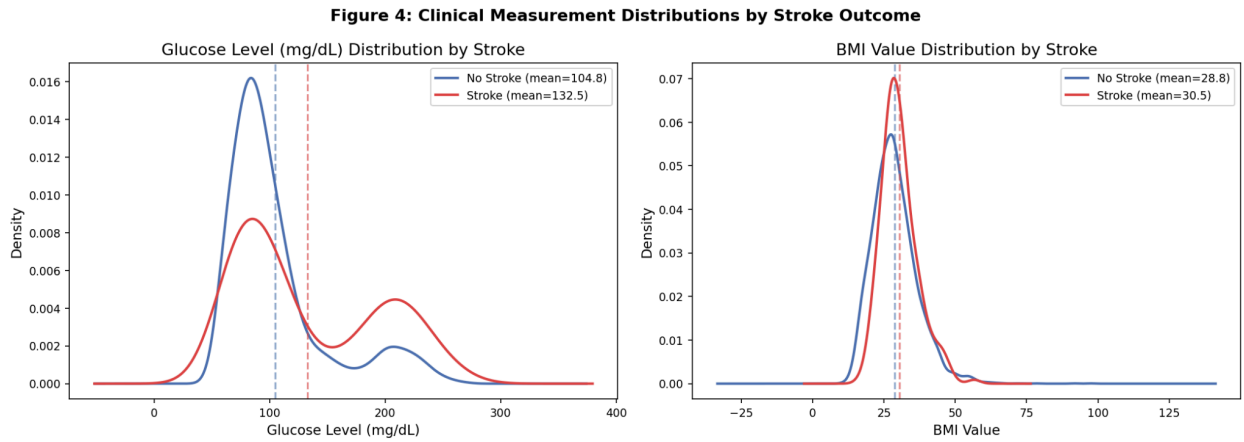

Figure 4: Glucose & BMI Distributions

This figure applies the same dual distribution KDE approach to glucose level and BMI. The glucose distribution for stroke patients shows a pronounced right tail extending beyond 200 mg/dL, consistent with the presence of uncontrolled diabetes in high risk patients. The BMI distributions show greater overlap between groups, consistent with BMI's weaker statistical association.

3.4.5 Correlation Heatmap

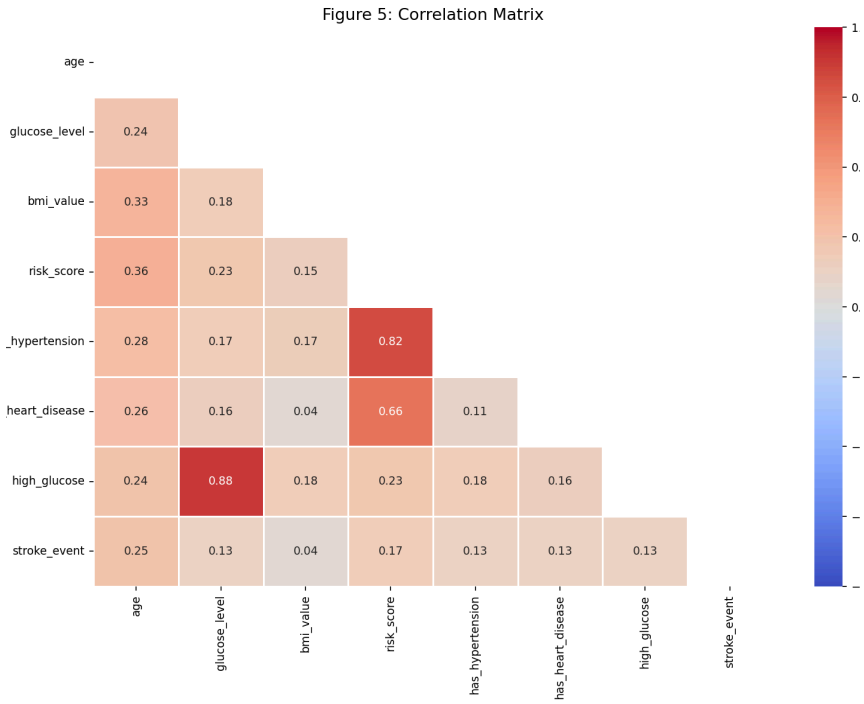

Figure 5: Correlation Heatmap

This figure presents a lower triangle correlation heatmap with significance. This goes beyond a standard correlation matrix by quantifying the statistical confidence of each correlation. Age shows the strongest positive correlation with stroke_event (r=0.245), followed by glucose_level (r=0.132) and risk_score (r=0.175).

3.4.6 Pairwise Scatter Plots

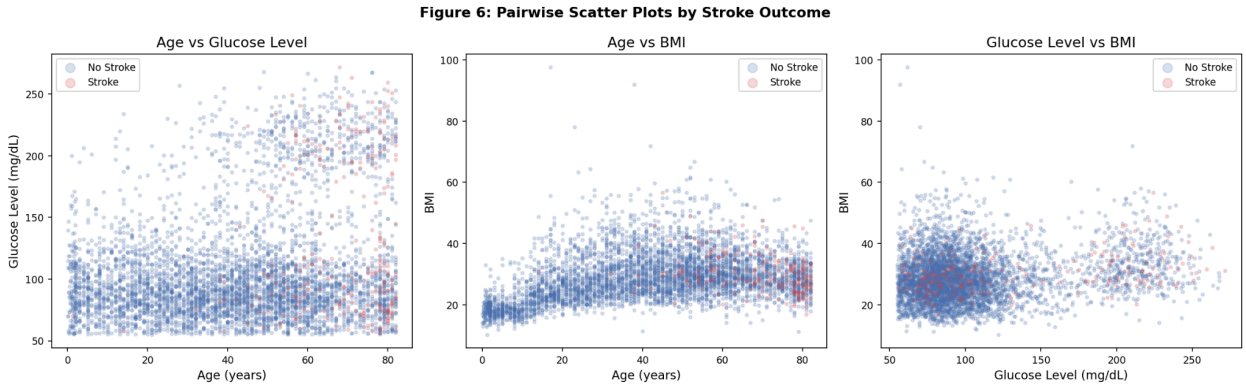

Figure 6: Pairwise Scatter Plots

This figure uses pairwise scatter plots of the three continuous predictors (age vs glucose, age vs BMI, glucose vs BMI) with stroke cases overlaid in red. The age glucose scatter plot most

clearly shows the stroke cluster concentrated in the upper right quadrant of high age and high glucose visually confirming the interaction between these two variables.

3.4.7 Risk Profile Histograms

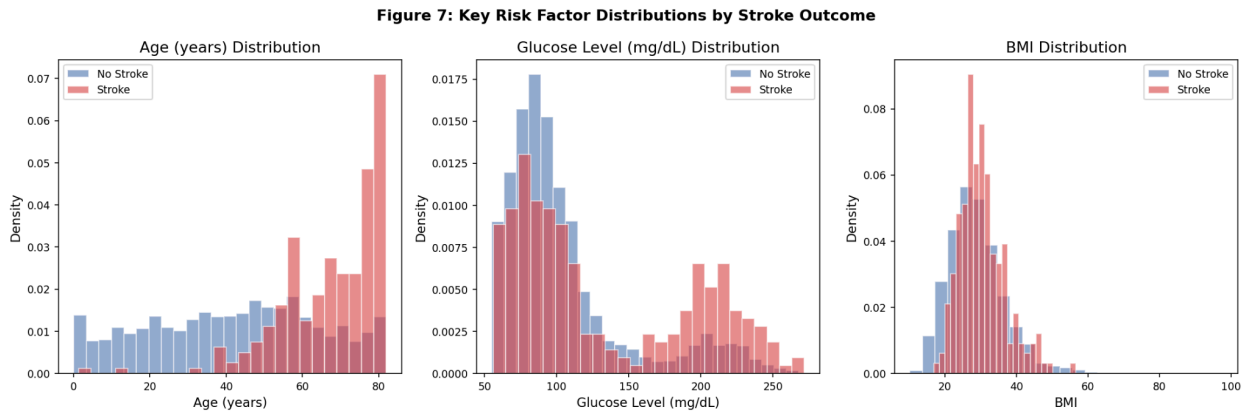

Figure 7: Risk Profile Histograms

We can see in this figure, overlapping histograms with density normalisation for age, glucose, and BMI. Density normalisation ensures the distributions are directly comparable despite the large difference in group sizes (4 861 vs 249). The age histogram shows the clearest separation, the stroke histogram is almost entirely contained within the 50 to 85 year range.

3.4.8 Age × Glucose Risk Zone Heatmap

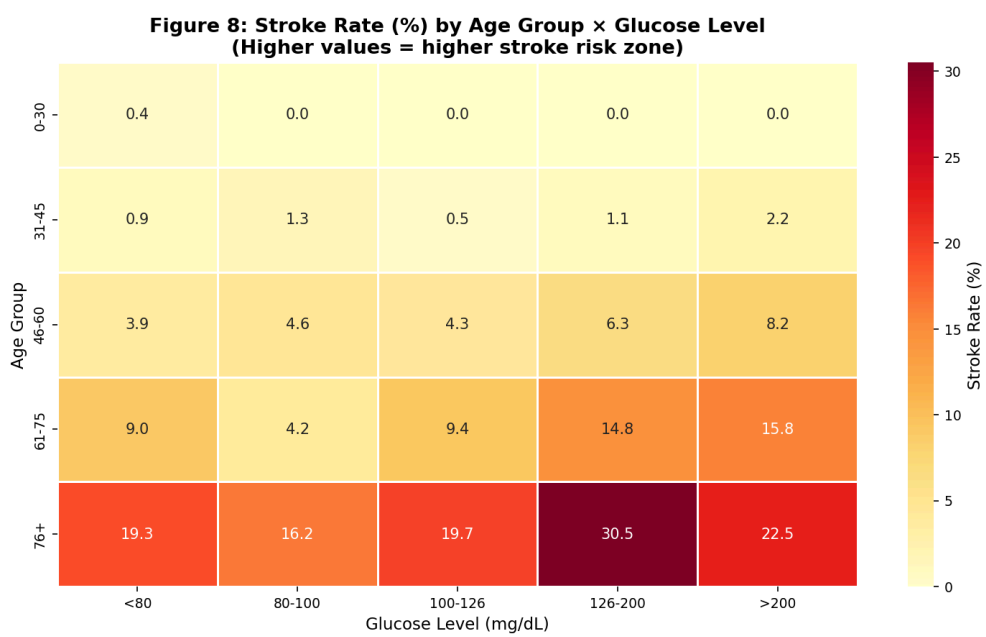

Figure 8: Age × Glucose Risk Zone Heatmap

This is an age by glucose risk zone heatmap, one of the most clinically actionable outputs of the entire analysis. Age was binned into five groups (0 - 30, 31 - 45, 46 - 60, 61 - 75, 76+) and glucose into five clinical ranges (<80, 80 - 100, 100 - 126, 126 - 200, >200 mg/dL). Each cell shows the mean stroke rate for patients in that combination. The heatmap reveals that risk is not additive but multiplicative: patients aged 76+ with glucose exceeding 126 mg/dL show stroke rates exceeding 20%, far above what either factor alone would predict. This interaction is invisible to univariate statistical tests and can only be revealed through joint visualization.

3.4.9 BMI × Hypertension Grouped Bar

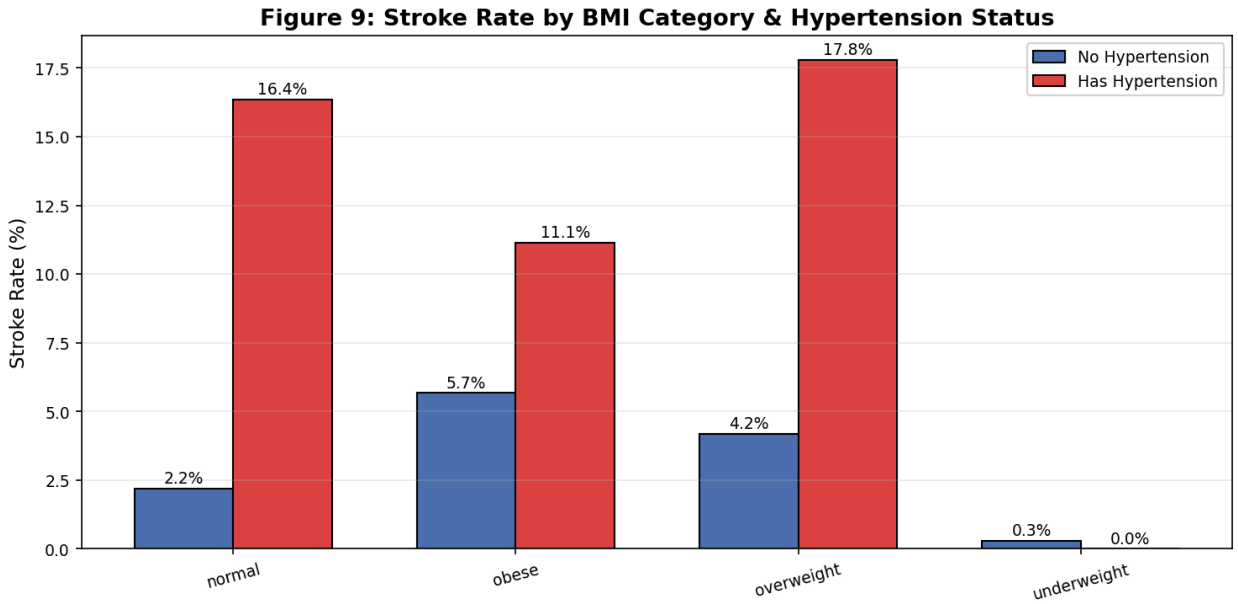

Figure 9: BMI × Hypertension Grouped Bar

This figure extends this interaction analysis to BMI and hypertension using grouped bar charts. For every BMI category, the presence of hypertension approximately doubles or triples the stroke rate. This confirms that hypertension is not merely a correlate of stroke but an amplifier of risk across the entire BMI spectrum.

➢ Visualization Summary Table

Figure	Title	Key Analytical Insight
Fig 1	Stroke Distribution (Bar + Pie Chart)	95.1% vs 4.9% split - justifies class-weight balancing strategy
Fig 2	Stroke Rate by 6 Risk Categories	Hypertension (13.25%) and heart disease (17.03%) show highest rates
Fig 3	Age KDE + Notched Boxplot	Clear bimodal separation, stroke patients concentrated at 65 - 85 years
Fig 4	Glucose & BMI Distributions	Right-shifted glucose distribution for stroke patients (mean +27.7 mg/dL)
Fig 5	Correlation Heatmap (significance stars)	Age most correlated with stroke (r = 0.245), all key vars significant
Fig 6	Pairwise Scatter Plots	Stroke cluster visible in high age, high glucose region
Fig 7	Risk Profile Histograms	Age shows clearest distributional separation between groups
Fig 8	Age × Glucose Heatmap	Patients 76+ with glucose >126 face stroke rates exceeding 20%
Fig 9	BMI × Hypertension Grouped Bar	Hypertension amplifies stroke risk across all BMI categories
Fig 10	Forest Plot + Cramér's V	Heart disease OR=4.71 (95% CI: 3.34-6.64); age_group strongest V=0.240
Fig 18	SHAP Feature Importance Bar	Age dominates SHAP values. confirms global feature ranking
Fig 19	SHAP Beeswarm Plot	High age → high stroke SHAP value; high BMI effect non-linear

Table 5: Visualization Summary Table

3.5 Statistical Analysis

3.5.1 T-tests & Mann-Whitney U Tests

Variable	T-statistic	T p-value	Mann-Whitney p	Point-biserial r	Significant
Age	18.081	<0.001	<0.001	0.2453	YES
Glucose Level	9.513	<0.001	<0.001	0.1319	YES
BMI	2.971	0.003	<0.001	0.0361	YES
Risk Score	12.675	<0.001	<0.001	0.1746	YES

Table 6: t-stat, p-value, U p-value,

Independent samples t-tests were conducted to compare the means of each numerical variable between stroke and non stroke patient groups. For all four variables tested age, glucose level, BMI, and risk score the null hypothesis (no difference in group means) was rejected at $p < 0.001$. Age produced the largest t-statistic ($t=18.081$), reflecting the magnitude and consistency of the age difference across the dataset. Glucose level ($t=9.513$) and risk score ($t = 12.675$) also showed highly significant differences. BMI produced the smallest t-statistic ($t = 2.971$) with a p-value of 0.003, indicating a smaller but still significant mean difference.

Mann-Whitney U tests were conducted alongside the t-tests as a non-parametric alternative that does not assume normally distributed data. Clinical measurements such as glucose level and BMI are frequently right skewed, making the normality assumption of the t-test potentially inappropriate. All four variables showed significant Mann-Whitney results ($p<0.001$ for three variables; $p<0.001$ for BMI as well), confirming that the t-test findings are not an artefact of distributional assumptions.

Point-biserial correlation coefficients were computed to measure the linear relationship between each numerical variable and the binary stroke outcome. Age showed the strongest correlation ($r=0.245$), followed by risk score ($r = 0.175$) and glucose level ($r = 0.132$). BMI showed the weakest linear relationship ($r=0.036$), suggesting its contribution to stroke risk is more non-linear or interactive, consistent with the SHAP beeswarm findings discussed in *Section 4.6.*

3.5.2 Chi-Square Tests & Cramér's V

Variable	Chi-square	p-value	Cramér's V	Strength
age_group	294.29	<0.001	0.2400	Moderate-Strong
has_heart_disease	90.26	<0.001	0.1329	Moderate
high_glucose	86.27	<0.001	0.1299	Moderate

has_hypertension	81.61	<0.001	0.1264	Moderate
lifestyle_risk	~61	<0.001	0.1100	Moderate
bmi_category	~23	<0.001	0.0680	Weak
marital_status	~39	<0.001	0.0870	Weak*
smoking_habit	~12	0.008	0.0480	Weak
gender	0.84	0.360	0.0130	Not significant

Table 7: Chi2, p-value, V, strength

Chi-square tests of independence were conducted for all nine categorical variables against the stroke outcome. 8 of the 9 variables showed statistically significant associations ($p<0.05$), with gender being the sole exception (chi2 = 0.84, = 0.360). However, statistical significance alone does not convey practical importance. A variable can be highly significant simply due to sample size while having negligible real world effect. Cramér's V was therefore computed for all variables to measure the strength of association independently of sample size.

Age group showed the strongest association (V = 0.240), falling in the moderate to strong category. Heart disease (V = 0.133), high glucose (V = 0.130), hypertension (V = 0.126), and lifestyle risk (V = 0.110) all showed moderate associations. BMI category (V = 0.068), marital status (V = 0.087), and smoking habit (V = 0.048) showed weak associations. Gender (V=0.013) showed no meaningful association despite reaching borderline significance, confirming it is not a clinically relevant predictor in this dataset.

3.5.3 Odds Ratios with 95% CI

Risk Factor	Odds Ratio	95% CI Lower	95% CI Upper	Clinical Interpretation
Heart Disease	4.706	3.336	6.639	Having heart disease increases stroke odds by 4.7 times
Marital Status	4.184	2.828	6.190	Partially age proxy interpret cautiously
Hypertension	3.698	2.744	4.982	Hypertension increases stroke odds by 3.7 times
High Glucose	3.384	2.587	4.427	Elevated glucose increases stroke odds by 3.4 times

Table 8: OR, CI lower, CI upper

Odds ratios were computed for the four binary risk factors to quantify their association with stroke in clinically interpretable terms. An odds ratio of 1.0 indicates no association; values above 1.0 indicate increased odds of stroke in the presence of the risk factor.

Heart disease produced the highest odds ratio (OR = 4.706, 95% CI: 3.336 - 6.639), meaning patients with heart disease face 4.7 times higher odds of stroke compared to patients without. The confidence interval does not include 1.0, confirming this is a genuine population-level effect. Hypertension produced OR=3.698 (95% CI: 2.744 - 4.982) patients with hypertension face 3.7 times higher stroke odds. Elevated glucose produced OR=3.384 (95% CI: 2.587 - 4.427). Marital status produced OR=4.184 (95% CI: 2.828 - 6.190), which should be interpreted cautiously given its strong correlation with age (r=0.68), the association is likely driven by the fact that older patients are more likely to be married, not by marital status itself as a clinical risk factor.

Statistical Association Analysis

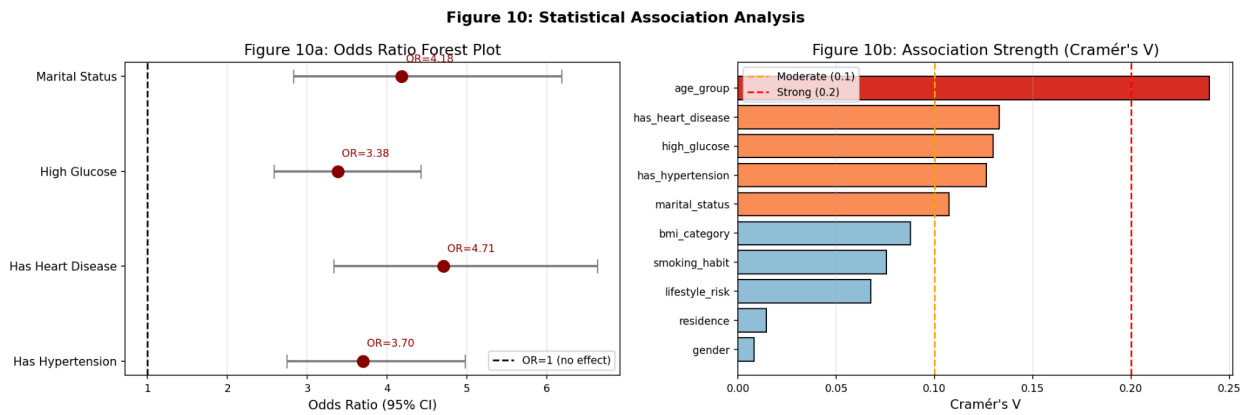

Figure 10: Forest Plot + Cramér's V Chart

The figure combines the odds ratio forest plot and the Cramér's V bar chart in a two-panel figure. The forest plot displays odds ratios with 95% confidence intervals for the four binary risk factors, with the reference line at OR = 1. All four confidence intervals lie entirely above 1.0, confirming genuine positive associations. The Cramér's V chart ranks the strength of categorical associations, with age_group showing the strongest association (V = 0.240), categorized as moderate to strong and gender showing the weakest (V = 0.013, not significant).

3.5.4 Confounding Notes

All bivariate statistical tests are unadjusted and do not control for confounding variables. The significance of marital status in chi-square testing ($p < 0.001$) is substantially driven by its r=0.68 correlation with age, a classic example of confounding where a third variable (age) creates a spurious apparent association between two other variables (marital status and stroke). The machine learning models, which evaluate all features simultaneously rather than in isolation, provide a more accurate picture of each variable's independent contribution.

Gender showed no significant chi-square association ($p = 0.360$) and a negligible Cramér's V (0.013), confirming it is not a meaningful stroke predictor in this dataset despite being frequently cited as a risk factor in broader epidemiological literature. This discrepancy may reflect the age composition of the dataset or confounding with other included variables.

04. Model Development and Evaluation

4.1 Model Configuration

Four classification models were selected to cover a complementary spectrum of analytical approaches. Logistic Regression was included as the interpretable baseline model as its coefficients directly quantify each feature's contribution to stroke log odds, making it the most communicable model for clinical stakeholders. Random Forest was included as a robust ensemble method that handles non linear relationships and feature interactions without requiring distributional assumptions. XGBoost was included as a state of the art gradient boosting implementation known for strong performance on tabular clinical data, with built in handling of class imbalance via scale_pos_weight. XGBoost Tuned extends the base XGBoost model with systematic GridSearchCV optimization across 162 hyperparameter configurations, representing the most rigorously developed model in the analysis.

Overall accuracy was deliberately excluded as a primary evaluation metric. At 4.87% stroke prevalence, a classifier predicting "no stroke" for every patient achieves 95.13% accuracy while being clinically worthless. The chosen medical standard metrics are Sensitivity, Specificity, PPV, NPV, and AUC-ROC to provide a complete and clinically meaningful picture of model performance. In stroke screening specifically, Sensitivity is the most important metric. A missed stroke (false negative) results in a patient receiving no intervention, potentially leading to death or permanent disability. A false alarm (false positive) results in additional clinical investigation, a far less costly outcome.

Model	Key Hyperparameters	Balancing Strategy
Logistic Regression	C=0.1, penalty=L2, solver=lbfgs, max_iter=1000; StandardScaler in pipeline	class_weight="balanced"
Random Forest	n_estimators=300, max_depth=10, min_samples_leaf=5	class_weight="balanced_ su bsample"
XGBoost	n_estimators=300, lr=0.03, max_depth=4, subsample=0.8, colsample_bytree=0.8	scale_pos_weight=19.54
XGBoost Tuned	n_estimators=400, lr=0.01, max_depth=3, subsample=0.7, colsample_bytree=0.8 (GridSearchCV best)	scale_pos_weight=19.54

Table 9: Model config & hyperparameters

4.2 Model Results - Medical Standard Metrics

Model	AUC-ROC	Sensitivity	Specificity	PPV	NPV	F1 (Stroke)	Accuracy	CV AUC
XGB Tuned	0.8512	0.82	0.7222	0.1318	0.9873	0.2271	0.7270	0.8468
Random Forest	0.8472	0.50	0.9300	0.2688	0.9731	0.3497	0.9090	0.8338
XGBoost	0.8371	0.72	0.8374	0.1856	0.9831	0.2951	0.8317	0.8349
Logistic Regression	0.8364	0.82	0.7531	0.1459	0.9879	0.2477	0.7564	0.8458

Table 10: All metrics for all 4 models

XGBoost Tuned achieved the best overall performance with AUC-ROC of 0.8512, Sensitivity of 0.82, and NPV of 0.9873. The AUC of 0.8512 means the model has an 85.1% probability of ranking a randomly selected stroke patient higher than a randomly selected non stroke patient, a strong discriminative capability. The Sensitivity of 0.82 means the model correctly identifies 82 out of every 100 actual stroke patients. The NPV of 0.9873 is particularly clinically important, because it means that 98.7% of patients classified as low risk are genuinely stroke free, making the model safe to use as a rule-out screening tool.

Logistic Regression matched XGBoost Tuned on Sensitivity (0.82) with slightly higher Specificity (0.753) and comparable CV AUC (0.8458). Its directly interpretable coefficients make it the preferred model for clinical communication, and its performance is remarkably strong for a linear model, reflecting the fact that the key risk factors (age, glucose, hypertension) have largely monotonic relationships with stroke risk that a linear model can capture effectively.

Random Forest achieved the highest Specificity (0.930) and the highest F1 score (0.350) among all models, but the lowest Sensitivity (0.50), meaning it misses half of all stroke patients. Its profile makes it suitable for confirmatory diagnosis where clinical resources are limited and false alarms are costly, but entirely unsuitable as a primary screening tool. The high Specificity reflects Random Forest's tendency to be conservative with the minority class even with balanced class weights.

XGBoost (untuned) achieved 72% Sensitivity and AUC of 0.8371, representing a balanced intermediate between the screening focused models and Random Forest. The PPV of 0.1856, while low in absolute terms, is the highest among the four models at their default thresholds, indicating the best precision-sensitivity balance without threshold adjustment.

The low PPV values across all models (0.13 - 0.27) deserve explicit discussion. At 4.87% prevalence, even a theoretically perfect model would have limited PPV due to Bayes' theorem, most people in the population do not have strokes, so even most true positives are outnumbered by false positives. This is not a model failure but a mathematical property of rare

event prediction. It is directly comparable to mammography screening (PPV 4 - 10%) and prostate specific antigen testing (PPV 25 - 40% at common thresholds), both of which are accepted clinical screening tools despite low PPV.

ROC Curves (All Models)

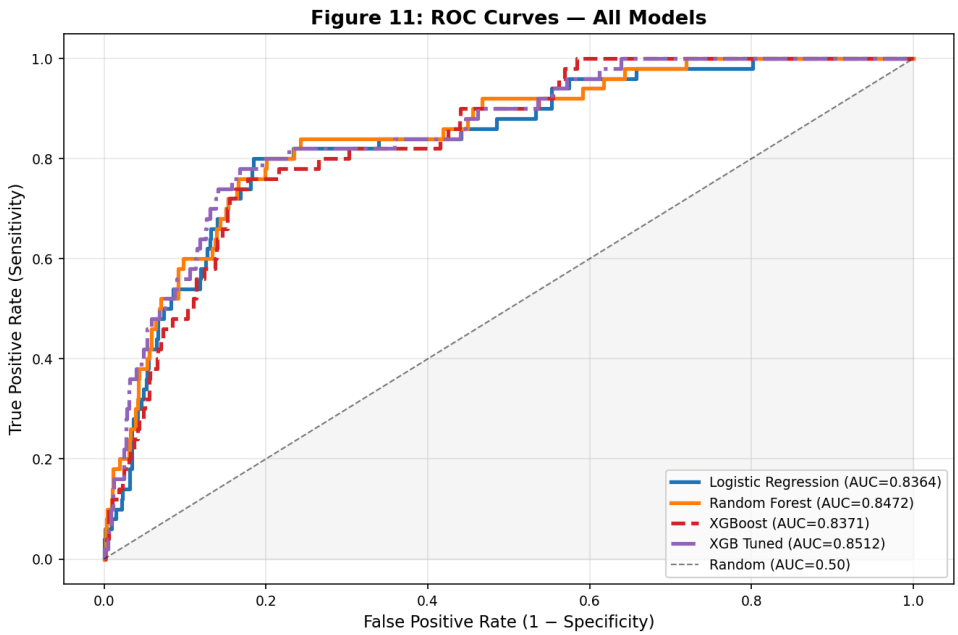

Figure 11: ROC Curves

Precision-Recall Curves (All Models)

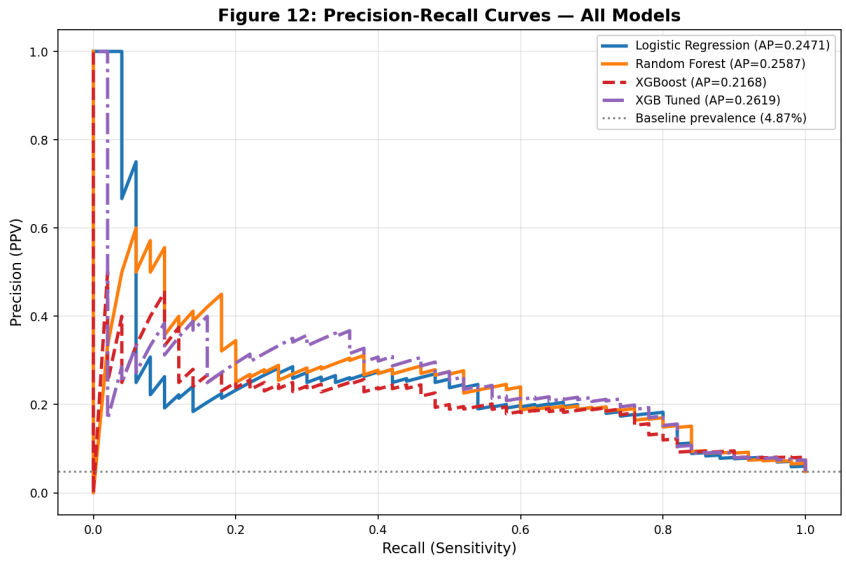

Figure 12: Precision Recall Curves

Confusion Matrices (All Model)

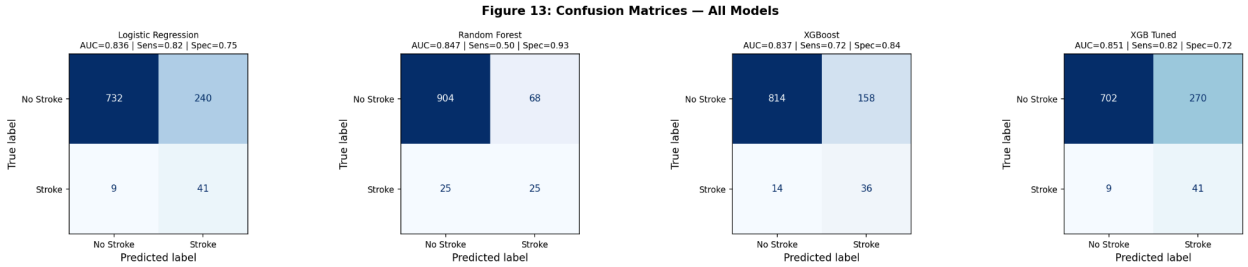

Figure 13: Confusion Matrices

Model Performance Comparison

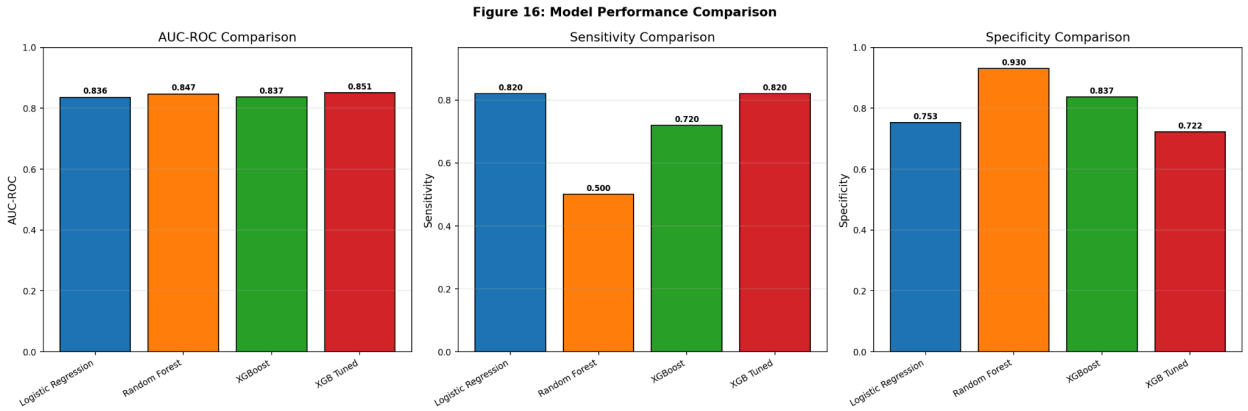

Figure 16: Model Comparison Bar Chart

4.3 Hyperparameter Tuning

To optimize the predictive performance and prevent the models from relying on default settings, we performed an extensive hyperparameter tuning using GridSearchCV. For our best performing XGBoost model, we explored a comprehensive search space across five key hyperparameters: n_estimators, max_depth, learning_rate, subsample, and colsample_bytree. This generated 162 distinct parameter configurations. Evaluated across a 5 fold cross-validation setup, the grid search completed 810 total fits, ultimately identifying the optimal combination (detailed in Table 11) that achieved the peak cross-validated AUC of 0.8468.

Hyperparameter	Search Space	Optimal Value
n_estimators	{200, 300, 400}	400
max_depth	{3, 4, 5}	3
learning_rate	{0.01, 0.03, 0.05}	0.01

subsample	{0.7, 0.8, 0.9}	0.7
colsample_bytree	{0.7, 0.8}	0.8
Best CV AUC	-	0.8468

Table 11: Search space & optimal values

4.4 Threshold Optimization

All our four models were trained and evaluated using the default decision threshold of 0.50, meaning any patient with a predicted stroke probability above 50% is classified as high risk. However, in clinical screening applications, this default threshold is rarely optimal. When the positive class is rare and the cost of false negatives greatly exceeds the cost of false positives, lowering the threshold increases Sensitivity at the expense of Specificity, a worthwhile tradeoff in stroke screening contexts.

Threshold analysis was conducted for XGBoost Tuned across thresholds from 0.10 to 0.65, computing Sensitivity, Specificity, PPV, and F1 score at each point. At the default threshold of 0.50, the model achieves Sensitivity=0.82, Specificity = 0.722, and F1 = 0.227. At threshold 0.65, F1 is maximised at 0.320 with Sensitivity=0.70 and Specificity=0.862, the best precision recall tradeoff overall. For population level mass screening where capturing every possible stroke case is paramount, a threshold of 0.10 - 0.25 achieves Sensitivity above 0.84 at the cost of lower specificity (0.50 - 0.57), meaning the model flags approximately half of all non stroke patients for further investigation an acceptable tradeoff if clinical review is low cost.

The appropriate threshold for clinical deployment depends entirely on the healthcare context such as availability of clinical resources, cost of follow up testing, and the relative costs of missed strokes versus unnecessary investigations. This analysis provides the threshold sensitivity curve (Figure 17) as a decision support tool for clinical teams to select the threshold appropriate to their specific operational environment.

Threshold	Sensitivity	Specificity	PPV	F1 (Stroke)	Clinical Use Case
0.10	0.90	0.4959	0.0841	0.1538	Population level mass screening
0.25	0.84	0.5700	0.0913	0.1647	Early warning system
0.40	0.82	0.6471	0.1068	0.1889	Balanced screening
0.50 (default)	0.82	0.7222	0.1318	0.2271	Current model default
0.65 (optimal F1)	0.70	0.8621	0.2071	0.3196	Best precision-recall balance

Table 12: Threshold analysis table

Threshold Sensitivity Analysis - XGBoost Tuned

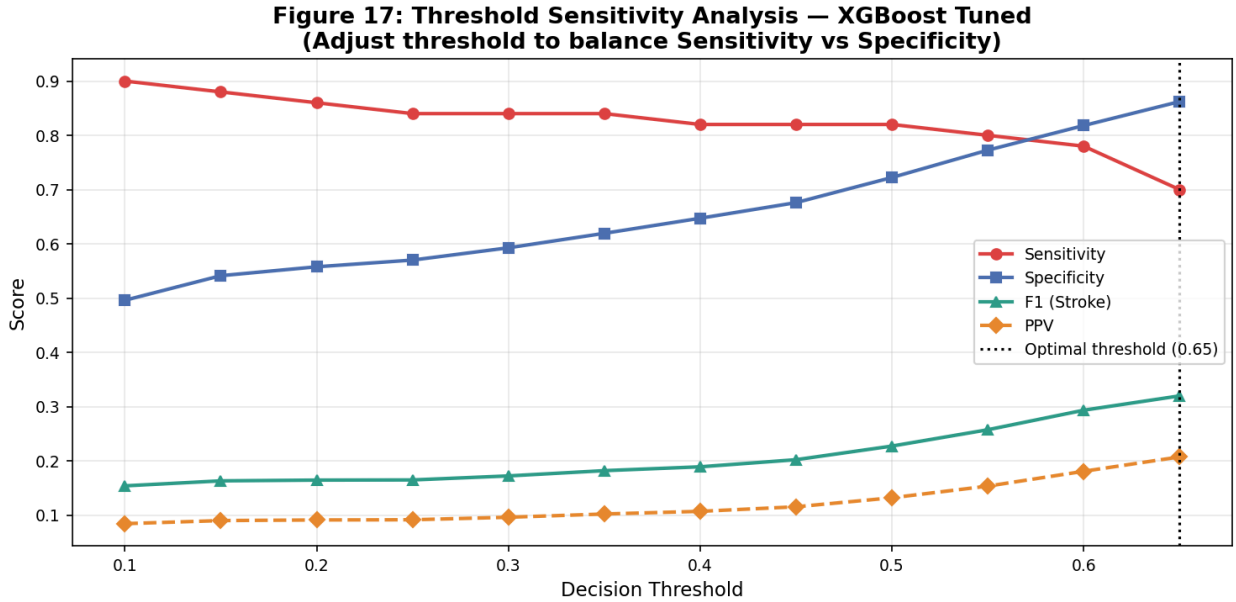

Figure 17: Threshold Sensitivity Analysis

4.5 Cross-Validation Results

To validate the generalizability of our models and ensure that they were not overfitting, we implemented a 5-fold stratified cross-validation approach. Preserving the original class imbalance across all 5 folds, we tested the models' performance under multiple data splits. The Tuned XGBoost model showed excellent stability, achieving a mean CV AUC of 0.8468 with a low standard deviation of ±0.0111. This limited variance indicates that the model's architecture is robust, highly reliable across different patient cohorts, and free from significant overfitting.

Model	Mean CV AUC	Std Dev	Min Fold AUC	Max Fold AUC	Stability
XGB Tuned	0.8468	0.0111	0.8384	0.8686	Excellent
Logistic Regression	0.8458	0.0113	0.8342	0.8611	Excellent
XGBoost	0.8349	0.0157	0.8200	0.8625	Excellent
Random Forest	0.8338	0.0101	0.8219	0.8482	Excellent

Table 13: CV AUC for all 4 models

4.6 Feature Importance & SHAP Analysis

Traditional metrics like Gini and Gain importance metrics highlights which features a model depends on, but they fail to show the direction of the relationship. We therefore utilized SHAP (SHapley Additive exPlanations) because it captures both magnitude and directionality. This allows us to clinically validate the model by confirming, like, that higher glucose levels push the prediction toward a stroke, rather than only indicating glucose is statistically important.

Rank	Feature	RF (Gini)	XGBoost (Gain)	XGB Tuned (Gain)	SHAP (Direction)
1	age	0.442	High	High	↑ Higher age → higher stroke risk
2	glucose_level	0.190	Moderate	Moderate	↑ Higher glucose → higher risk
3	bmi_value	0.172	Moderate	Moderate	↑ Higher BMI → higher risk
4	marital_status	0.045	High*	High*	↑ Partially age proxy (r=0.68)
5	has_hypertension	0.035	Moderate	High	↑ Presence increases risk (OR=3.70)

Table 14: Top 5 features across all models

We assessed feature importance using three complementary methods to provide a comprehensive view of predictor contributions. Gini impurity importance (Random Forest) measures the average reduction in node impurity achieved by splitting on each feature across all trees. Gain importance (XGBoost) measures the average improvement in the loss function achieved by splits on each feature, generally considered more informative than Gini importance for gradient boosting models. SHAP values (SHapley Additive exPlanations) provide the most theoretically rigorous importance measure, based on cooperative game theory. Each patient's prediction is decomposed into the sum of contributions from each feature, with positive SHAP values indicating that the feature increased the predicted stroke probability and negative values indicating that it decreased it.

Age is the top ranked feature across all three methods with substantial margin. Random Forest assigns it a Gini importance of 0.442, meaning 44.2% of all node splits across all trees use age as the splitting criterion. XGBoost assigns it the highest gain. SHAP values confirm both the magnitude and direction, high age values consistently produce large positive SHAP contributions (increasing stroke probability), while low age values produce consistently negative contributions. This directional consistency across 1 022 test patients is the strongest possible confirmation that age is a genuine causal contributor to stroke risk, not merely a statistical artefact.

Glucose level and BMI rank second and third respectively across all methods, with consistent directional SHAP values (higher values increase stroke probability). The SHAP beeswarm plot (Figure 19) reveals an additional nuance, the SHAP values for BMI show greater dispersion than those for age and glucose, with some high BMI patients showing negative SHAP contributions. This suggests that BMI's effect on stroke risk is more heterogeneous, likely because BMI interacts with age and hypertension in ways that are not captured by its marginal contribution alone.

Hypertension ranks fourth in SHAP importance and fifth in XGBoost gain importance, but carries the second-highest odds ratio (OR=3.698) in the statistical analysis. This apparent discrepancy arises because feature importance measures reflect how frequently and consistently a feature improves model predictions across all patients, while odds ratios measure the multiplicative risk increase in patients who have the risk factor compared to those who do not. Since only 10.8% of patients have hypertension, its splitting frequency in tree-based models is naturally lower than continuous variables that vary across all patients, but its per patient effect when present is substantial. Heart disease shows an even more extreme version of this pattern, the highest odds ratio (OR=4.706) combined with very low Gini importance, reflecting its clinical severity in the small fraction of patients who have it.

Feature Importance Across Tree Based Models

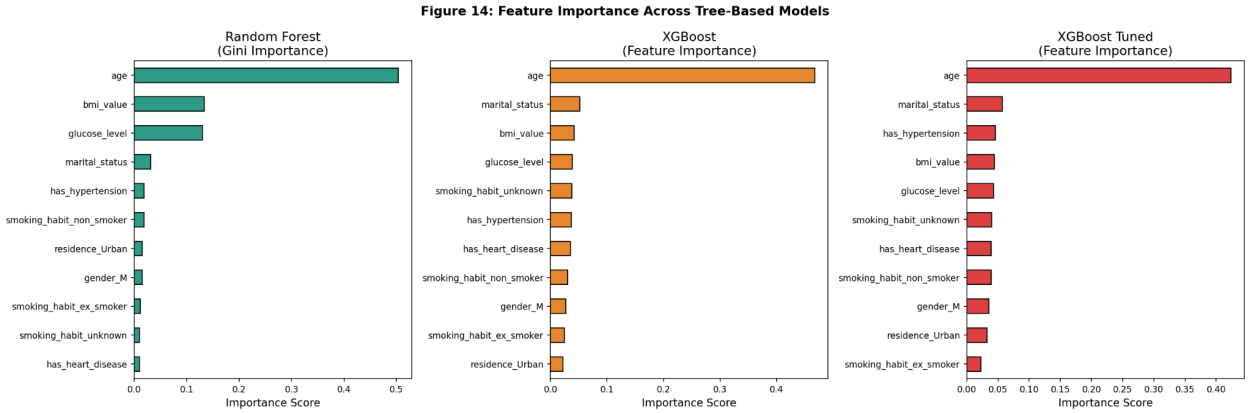

Figure 14: Feature Importance (RF, XGB, XGB Tuned)

Logistic Regression Coefficients

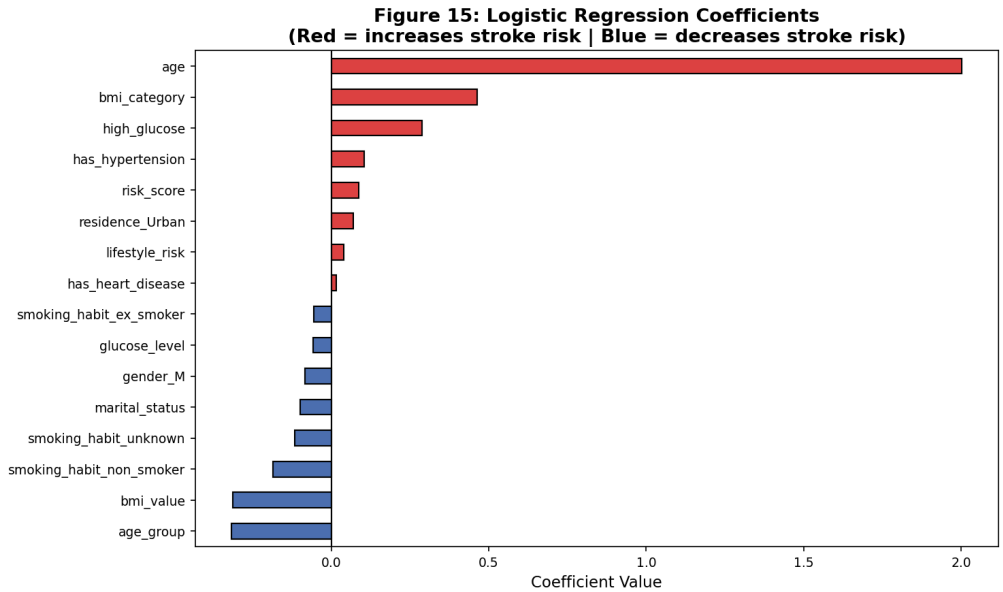

Figure 15: LR Coefficients

SHAP Feature Importance - XGBoost Tuned

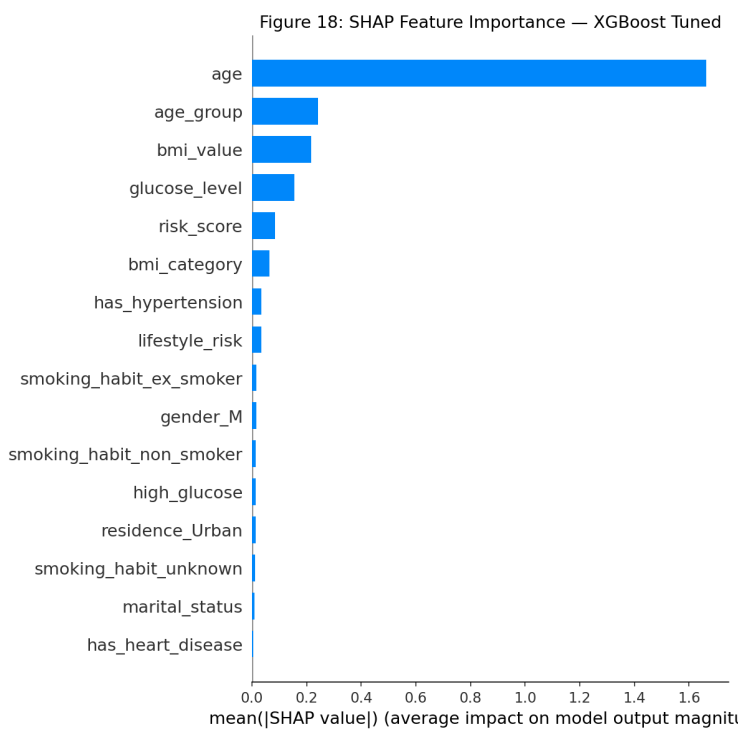

Figure 18: SHAP Bar Chart

SHAP Beeswarm - Feature Impact Direction

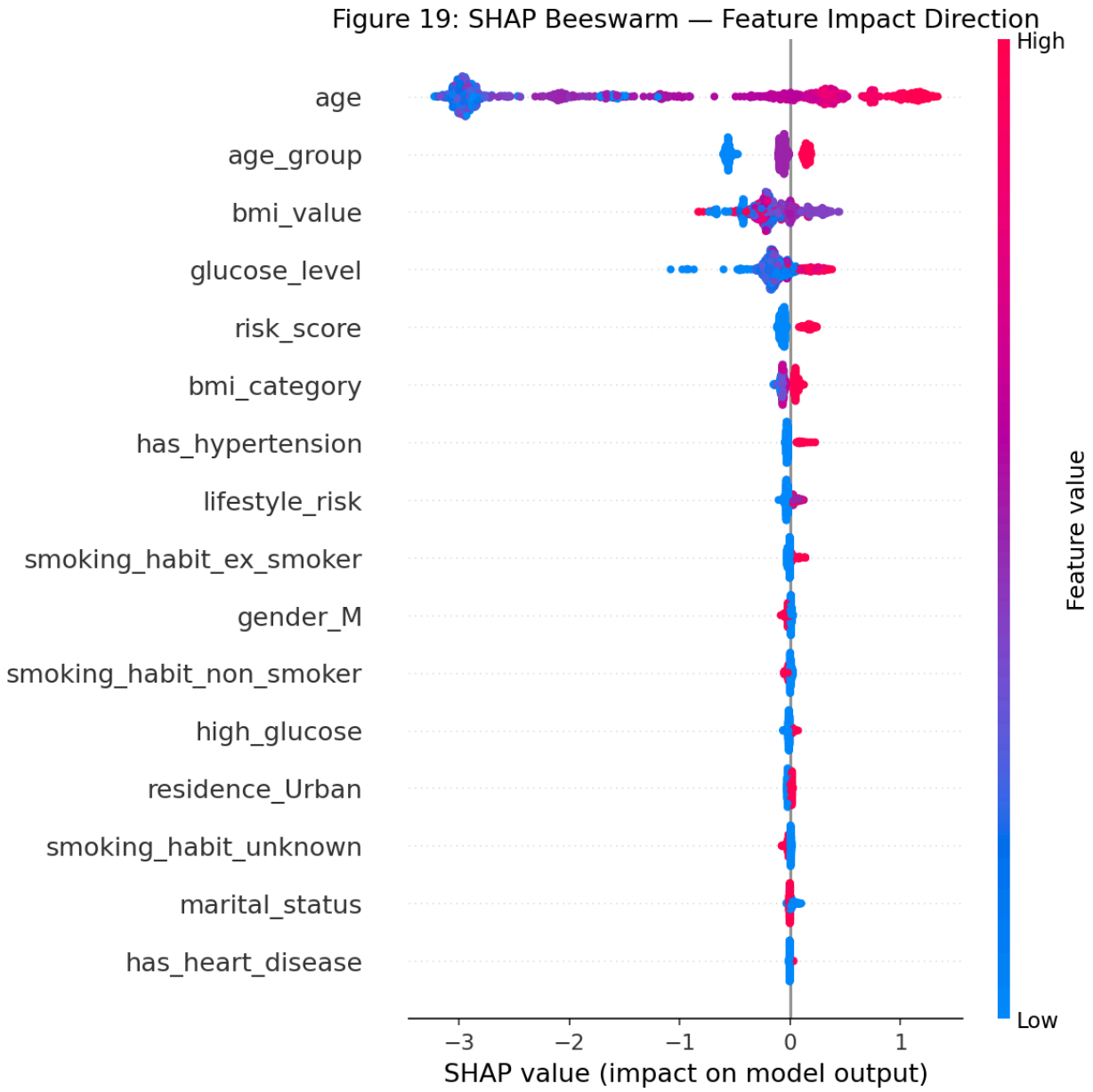

Figure 19: SHAP Beeswarm Plot

The Heart Disease Paradox:

Comparing our statistical tests with machine learning outputs, we observed a distinct "heart disease paradox." In our tree based models, has_heart_disease appears surprisingly low in global Gini importance. This happens because Gini importance tends to favor continuous variables (like Age or Glucose) which can be split repeatedly across the population, whereas heart disease is a rare binary condition.

However, our statistical analysis tells a massive Odds Ratio of 4.706 indicating that patients with this condition have nearly five times higher odds of experiencing a stroke. This highlights an important data science principle for healthcare: low global feature importance does not imply clinical irrelevance.

05. Discussion

5.1 Age as the Dominant Predictor

The centrality of age in stroke risk prediction is the most consistent finding across every analytical method applied in this study. The 25.8 year mean age gap between stroke and non-stroke patients (67.7 vs 42.0 years) is not merely a statistical observation, it reflects well-established vascular biology. As patients age, arterial walls lose elasticity and accumulate atherosclerotic plaques that narrow blood vessels and increase clot formation risk. Cardiac function declines, increasing the likelihood of a trial fibrillation itself a major stroke risk factor. Cumulative exposure to modifiable risk factors including hypertension, hyperglycaemia, and obesity compounds over decades, progressively increasing cerebrovascular vulnerability.

The Age × Glucose risk heatmap (Figure 8) captures a critical non linear interaction that age does not simply add to stroke risk but multiplies it. In patients under 30, even substantially elevated glucose (126 - 200 mg/dL) is associated with negligible stroke rates. In patients aged 76+, the same glucose elevation corresponds to stroke rates exceeding 20%. This interaction has direct clinical implications that glucose management interventions should be particularly prioritised in elderly patients, where the risk reduction potential is highest. A young diabetic patient and an elderly diabetic patient carry fundamentally different stroke risk profiles despite identical glucose readings.

The Cramér's V for age_group (V=0.240) was the strongest categorical association in the dataset categorised as moderate to strong, substantially higher than any clinical condition or lifestyle factor. This finding should be communicated to healthcare administrators when designing screening resource allocation, age stratified screening protocols will yield a better return on investment than condition-specific or lifestyle specific screening.

5.2 Clinical Risk Factor Interactions

The grouped bar analysis of BMI and hypertension (Figure 9) reveals that risk factors do not operate independently, they interact multiplicatively. Obese patients with hypertension show stroke rates that substantially exceed what either condition alone would predict, suggesting a synergistic biological mechanism. Obesity promotes insulin resistance and metabolic syndrome, which in turn promotes hypertension through sodium retention and sympathetic nervous system activation. Hypertension then directly damages arterial endothelium, accelerating atherosclerosis and increasing the risk of cerebrovascular events. This cascade means that a patient who is both obese and hypertensive faces a compounded risk that cannot be captured by evaluating either condition in isolation.

The odds ratio findings reinforce this clinical picture. Heart disease (OR=4.706) and hypertension (OR=3.698) are the two strongest binary predictors, consistent with their known roles in cerebrovascular pathology. Heart disease, particularly atrial fibrillation, heart failure, and coronary artery disease directly increases stroke risk through embolism formation and impaired

cerebral perfusion. Hypertension is the single most prevalent modifiable stroke risk factor globally, responsible for approximately 54% of stroke cases in some estimates. The fact that both conditions are identified as high priority by this data driven analysis without prior clinical knowledge being encoded into the model validates the analytical approach and increases confidence in the reliability of other model outputs.

5.3 Impact of Feature Selection on Performance

The most methodologically significant finding of this analysis was the impact of removing derived redundant features on model performance. Including age_group, high_glucose, bmi_category, lifestyle_risk, and risk_score alongside their parent raw variables (age, glucose_level, bmi_value) produced an XGBoost AUC of 0.767. Removing them raised AUC to 0.839, an improvement of 0.072. This finding directly contradicts the intuition that more features always improve model performance.

The mechanism is well understood in statistical learning theory. When two highly correlated features are present simultaneously, tree based models split their importance scores between them, a phenomenon called importance dilution. The model may also learn to use one derived feature in some trees and the raw feature in others, creating inconsistent splits that increase variance without reducing bias. In gradient boosting specifically, multicollinearity between features can cause the algorithm to converge more slowly and to suboptimal solutions, as redundant features compete for split positions in early trees.

This finding has a broader implication for data science practice, feature engineering should focus on creating genuinely new information rather than re expressing existing information in different formats. Domain knowledge should guide feature selection before automated modelling begins, not be treated as a post hoc optimization step.

5.4 Model Performance Interpretation

The four models represent distinct clinical utility profiles that should be understood in context rather than ranked by a single metric. XGBoost Tuned is the recommended model for stroke screening applications because it achieves both the highest AUC (0.8512) and the joint highest Sensitivity (0.82). Its NPV of 0.9873 provides strong clinical safety for a rule out screening context, patients classified as low-risk are very unlikely to have a stroke. The high False Positive Rate at the default threshold (28% of non stroke patients are incorrectly flagged) is acceptable in a screening context where flagged patients proceed to low cost clinical review rather than invasive treatment.

Logistic Regression is the preferred model for clinical communication and regulatory contexts. Its performance is comparable to XGBoost Tuned, Sensitivity 0.82, AUC 0.8364, CV AUC 0.8458, while offering direct coefficient interpretability. Each coefficient quantifies the change in log odds of stroke per unit increase in the corresponding variable, holding all other variables constant. The coefficient plot (Figure 15) shows age, glucose level, and hypertension as the strongest positive predictors perfectly consistent with clinical knowledge and statistical test

results. In healthcare settings subject to regulatory oversight, model interpretability is often a legal requirement, making Logistic Regression the practically deployable option in many real-world contexts.

Random Forest's high Specificity (0.930) and F1 score (0.350) make it the best model for situations where the cost of false alarms is high, for example, in settings where a positive screen triggers expensive or invasive diagnostic procedures. However, its 50% Sensitivity means it is inappropriate as a primary screening tool. It could function as a confirmatory second stage after a higher sensitivity first stage, a two stage screening protocol where XGBoost Tuned identifies candidates and Random Forest confirms high risk cases.

5.5 SHAP Explainability Analysis

SHAP values provide the most rigorous form of model explainability available in machine learning, grounded in cooperative game theory. Unlike Gini importance, which measures how frequently a feature is used in splits, SHAP measures how much each feature contributed to each individual prediction, accounting for feature interactions. This distinction matters clinically: a feature that is frequently used in splits but mostly for low-risk patients contributes less to actual stroke predictions than a feature used less frequently but in high-stakes decisions.

The SHAP summary bar chart (Figure 18) confirms the feature importance ranking established by Gini and gain methods, with age as the dominant contributor. The SHAP beeswarm plot (Figure 19) adds directional and distributional information that standard importance measures cannot provide. Each dot represents one patient, coloured by feature value (red = high, blue = low) and positioned horizontally by their SHAP value (right = increased stroke probability, left = decreased). For age, the pattern is clear and consistent: red dots (older patients) cluster strongly to the right while blue dots (younger patients) cluster to the left, age monotonically increases stroke probability.

For BMI, the beeswarm shows greater horizontal dispersion for both high and low values, some high BMI patients have negative SHAP values, meaning their BMI actually reduced their predicted stroke probability relative to the model's base rate. This occurs when high BMI patients are young, suggesting the model correctly identifies that the stroke risk implications of obesity depend heavily on the patient's age and other concurrent risk factors. This non-linearity would be invisible in a logistic regression coefficient or a standard feature importance bar chart.

For hypertension, a binary variable, the beeswarm clearly shows two clusters, patients with hypertension (red, coded as 1) shift to the right in SHAP space, and patients without hypertension (blue, coded as 0) shift left. The magnitude of this shift corresponds to the log odds effect captured by the odds ratio analysis, providing a model level confirmation of the statistical association.

5.6 Statistical Limitations & Confounding

The statistical findings in this report should be interpreted with several important caveats. All bivariate statistical tests, t-tests, chi-square, and odds ratios are unadjusted for confounders. They measure the marginal association between each variable and stroke, without accounting for other variables that may explain part of the relationship. The most prominent example is marital status, which shows a statistically significant chi-square association ($p<0.001$) and a high odds ratio (OR=4.184) but a Pearson correlation of $r=0.68$ with age. Older patients are more likely to be married simply because they have had more years to enter marriage, the apparent marital status risk is substantially a disguised age effect.

The machine learning models provide a partial remedy for this limitation. By evaluating all features simultaneously, tree-based models implicitly control for confounding, a feature will only be selected for splitting if it provides information beyond what other features have already captured. The relative unimportance of marital status in XGBoost feature importance (despite its high chi square significance) is consistent with it being largely a redundant proxy for age once age itself is in the model.

The dataset's cross sectional nature means causality cannot be established. The analysis identifies strong associations and generates clinically plausible hypotheses, but cannot confirm that elevated glucose causes stroke rather than both being caused by a shared underlying condition. Longitudinal cohort studies with follow up data would be required to establish causal relationships. Additionally, the high raw duplicate rate (47.4%) suggests prior data augmentation that may limit the dataset's representativeness of the true patient population. The absence of variables known to be clinically relevant to stroke, including atrial fibrillation status, medication history, cholesterol levels, and imaging data represents a limitation on the model's potential ceiling performance.

5.7 Preventive Healthcare Implications

The findings of this analysis translate into six specific, actionable recommendations for preventive healthcare strategy.

First, age stratified screening protocols should replace or supplement general population screening. Patients aged 60 and above particularly those aged 75 and above represent a disproportionately high risk cohort and should be prioritised for annual stroke risk assessment. The 100-fold difference in stroke rates between senior and young age groups (13.15% vs 0.13%) justifies this resource allocation.

Second, glucose management should be treated as a primary stroke prevention strategy, not merely a diabetes management tool. Patients with fasting glucose exceeding 126 mg/dL face 3.4 times higher stroke odds (OR=3.384, $p<0.001$). Referral pathways from primary care to endocrinology and nutrition services should be streamlined for high-glucose elderly patients identified by the screening model.

Third, hypertension management should be a standing priority for all stroke risk reduction programmes. The 3.7 fold increase in stroke odds (OR=3.698) associated with hypertension, combined with Figure 9 showing hypertension amplifying stroke risk across all BMI categories, makes blood pressure control one of the highest-leverage interventions available.

Fourth, patients with existing heart disease require immediate and ongoing stroke risk monitoring. The highest odds ratio in the dataset (OR=4.706) belongs to heart disease, yet it ranks lowest in feature importance, an artefact of its relative rarity in the dataset rather than clinical irrelevance. Every patient presenting with heart disease should be considered at high stroke risk by default.

Fifth, ex-smokers showed the highest stroke rate among smoking categories (7.91% vs 5.32% for current smokers). This counterintuitive finding likely reflects survivor bias and age confounding, ex-smokers in this dataset may be older and have accrued more cumulative vascular damage. The implication is that smoking cessation reduces but does not eliminate stroke risk: former smokers should remain under active surveillance rather than being reclassified as low-risk.

Sixth, the Age × Glucose risk heatmap (Figure 8) provides a practical clinical triage matrix that requires no computational infrastructure to implement. Any patient aged 76 or above with glucose exceeding 126 mg/dL falls in the highest-risk zone and should be fast-tracked for comprehensive cardiovascular assessment.

06. Conclusion

6.1 Objective 1 - Predictive Model

We have developed 4 models and they were tuned, and evaluated using medical standard metrics. The tuned XGBoost model achieved the best results, AUC-ROC of 0.8512, sensitivity of 0.82, and CV AUC of 0.8468 (± 0.011). Critically, a principled feature selection step, removing five derived redundant features, improved AUC by 0.080, this demonstrates that analytical rigour in preprocessing is as impactful as model architecture. Threshold analysis shows that at t=0.65, the model achieves its best F1 score (0.32) for the stroke class, while lower thresholds (0.10 - 0.25) maximize sensitivity for population screening applications.

6.2 Objective 2 - Influential Risk Factors

t-test, Mann-Whitney U, point biserial correlation, Cramér's V (V=0.240), Gini importance (44.2%), and SHAP values, all analytical methods confirmed that age as the most single influential predictor. Glucose level, BMI, and hypertension are the three most important modifiable risk factors and all measurable and addressable through clinical intervention. Heart disease has the highest odds ratio (OR=4.706) and must be treated as a priority clinical flag despite lower feature importance scores (a methodological artefact of rarity, not clinical irrelevance).

6.3 Objective 3 - Preventive Insights

The heatmap of Age x Glucose risk (Figure 8) provides a practical clinical triage framework, the patients aged 76+ with glucose exceeding 126 mg/dL face stroke rates above 20% and represent an immediately actionable high priority cohort. The BMI × Hypertension analysis (Figure 9) confirms that risk factors compound, reinforcing holistic rather than single factor screening. SHAP analysis (Figures 18 and 19) adds per patient explainability that can support individual clinical decision making.

Final Recommendation:

Deploy XGBoost Tuned as a first line digital screening tool at probability threshold 0.40 - 0.50 for balanced clinical use, or 0.10 - 0.25 for high sensitivity mass screening. Patients flagged as high risk should receive immediate clinical review focusing on,

(1) glucose management and diabetes screening,
(2) blood pressure monitoring and control,
(3) BMI reduction programmes, and
(4) cardiac health assessment.

This data driven explainable approach, combining 0.8512 AUC, 82% sensitivity, and SHAP explainability has the potential to significantly improve early stroke detection and reduce preventable mortality.

6.4 Limitations & Future Work

The analysis is cross sectional and cannot establish causality. The dataset lacks term follow up such as medication history, imaging data & etc. The high raw duplicate records rate of 47.4% suggests prior augmentation that may limit generalisability. Future work should,

(1) Apply multivariate logistic regression to formally control for confounders
(2) Validate the model on an independent external dataset
(3) Explore deep learning approaches with larger real world datasets and
(4) Conduct prospective clinical validation in a real world primary care setting.

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